

GERALDTON, Australia

WMO: 944030

Lat: 28.8044S

Lon: 114.6989E

Elev: 99

StdP: 14.64

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.5	43.7	21.1	15.9	75.9	26.3	20.3	74.6	27.3	63.1	24.8	63.4	7.2	130	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	21.2	100.9	67.1	97.0	66.5	93.1	66.3	74.4	85.6	72.9	83.4	71.6	81.7	15.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.5	116.9	77.9	69.9	110.3	77.1	68.4	104.9	76.1	38.0	85.5	36.6	83.6	35.5	81.7	92.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.2	25.0	23.0	DB	37.3	108.8	1.8	2.7	36.0	110.7	34.9	112.3	33.9	113.8	32.6	115.8	
			WB	35.9	77.5	1.8	3.7	34.6	80.2	33.6	82.4	32.6	84.5	31.3	87.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.6	76.2	78.1	76.2	71.0	65.8	60.9	58.2	58.2	59.9	64.3	69.6	73.5
	DBStd	8.65	5.48	6.23	6.03	5.03	4.41	3.83	3.79	3.77	4.07	4.80	5.42	5.89
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD65	848	0	0	0	5	41	130	214	213	163	70	11	1
	CDD50	6424	813	787	811	631	491	328	254	255	296	443	587	728
	CDD65	1797	348	367	347	187	67	8	2	3	9	48	147	264
	CDH74	16318	3126	3657	3179	1390	448	52	11	36	147	523	1349	2400
	CDH80	7172	1436	1789	1450	489	96	3	0	5	36	183	581	1104
Wind	WSAvg	12.2	15.2	14.5	13.4	11.1	10.5	10.5	10.2	10.0	10.5	12.0	13.8	15.1
Precipitation	PrecAvg	17.2	0.2	0.3	0.6	0.9	2.6	3.7	3.5	2.5	1.3	0.7	0.4	0.2
	PrecMax	29.7	2.5	3.3	5.3	3.0	11.1	10.1	7.3	6.1	3.0	4.0	1.9	0.9
	PrecMin	7.7	0.0	0.0	0.0	0.0	0.4	0.2	0.9	0.3	0.2	0.0	0.0	0.0
	PrecStd	4.4	0.4	0.7	1.0	0.8	2.0	2.1	1.5	1.0	0.7	0.7	0.5	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.8	105.6	102.4	95.0	87.5	78.9	75.2	78.8	84.7	92.8	100.0	104.2
		MCWB	68.3	68.0	67.5	63.1	60.4	55.2	57.6	57.4	59.1	61.3	63.4	66.7
	2%	DB	98.9	100.0	97.1	89.9	82.5	75.1	71.3	73.0	78.1	84.9	93.0	97.8
		MCWB	67.7	68.3	66.8	62.8	59.7	56.7	58.0	59.3	58.4	60.0	61.8	65.5
	5%	DB	93.1	95.6	92.7	85.6	78.8	71.9	68.6	70.0	73.3	79.2	86.6	91.3
		MCWB	67.9	68.6	66.1	63.7	59.8	57.9	58.1	59.2	59.7	60.5	62.0	65.1
	10%	DB	87.6	90.8	88.0	81.7	75.5	69.5	66.5	67.6	70.1	75.1	80.9	85.6
		MCWB	67.8	68.7	66.4	64.2	60.5	57.8	57.6	58.3	59.1	61.0	63.2	65.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.9	76.9	74.6	72.0	69.2	65.5	65.7	65.6	65.5	67.3	69.8	73.1
		MCDB	86.3	88.5	83.3	78.4	73.5	69.5	68.6	70.3	72.5	76.7	79.3	86.3
	2%	WB	73.8	74.9	72.9	70.3	67.3	63.6	62.5	63.1	63.6	65.5	67.9	70.9
		MCDB	84.9	87.0	82.6	76.8	72.7	68.3	66.5	68.2	70.7	74.3	77.7	82.5
	5%	WB	72.1	73.4	71.7	69.0	65.4	61.9	60.6	61.2	61.9	64.2	66.6	69.7
		MCDB	82.9	84.3	81.1	76.0	71.9	67.0	65.1	66.8	69.3	73.0	77.0	81.1
	10%	WB	70.5	71.9	70.5	67.7	63.6	60.2	59.1	59.5	60.4	63.0	65.4	68.3
		MCDB	81.1	82.3	79.9	75.5	71.1	65.8	64.3	65.4	67.7	71.5	75.8	79.5
Mean Daily Temperature Range	5% DB	MDBR	20.8	21.2	21.5	21.3	19.9	18.0	17.9	19.8	22.3	23.0	21.9	21.2
		MCDBR	32.0	29.1	28.5	27.1	24.6	21.9	21.2	23.9	29.0	32.1	33.7	33.5
		MCWBR	10.2	9.3	10.4	11.3	11.5	11.3	12.5	13.8	14.5	13.9	11.7	11.4
	5% WB	MCDBR	20.5	20.3	19.4	19.3	17.8	15.7	16.1	19.4	22.8	23.1	22.6	22.2
		MCWBR	7.6	7.5	8.1	9.7	10.5	10.3	11.4	12.8	13.2	11.6	9.7	8.9
Clear-Sky Solar Irradiance	taub		0.352	0.353	0.337	0.323	0.305	0.295	0.295	0.301	0.318	0.324	0.330	0.337
	taud		2.548	2.574	2.596	2.619	2.637	2.649	2.634	2.612	2.542	2.547	2.549	2.556
	Ebn at Noon		314	308	303	293	284	280	285	296	304	314	318	319
	Edh at Noon		35	33	30	27	24	23	23	27	31	33	35	35
All-Sky Solar Radiation	RadAvg		2586	2371	2005	1556	1209	1013	1077	1384	1796	2227	2529	2686
	RadStd		137	97	121	78	69	64	56	40	79	76	68	97

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page